

Chapter 56: Renal and Urinary Tract Disorders —

Revision Notes

Williams Obstetrics 26e, pp. 2584–2629 of the PDF. High-yield numbers are in **bold**. The six tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 6 figures are included from the PDF.

Big picture: Pregnancy dilates the collecting system (right > left) and raises GFR **~65%**, so creatinine normally falls and infections ascend easily. **Screen and treat asymptomatic bacteriuria** to prevent pyelonephritis — the **most common serious nonobstetric medical complication** and the leading cause of septic shock in pregnancy. For chronic renal disease, outcome depends on **degree of renal insufficiency and hypertension**, not on the specific lesion.

1. Pregnancy-Induced Urinary Tract Changes

- Kidney disorders in **~0.3%** of births (Australia).
- Kidneys enlarge; calyces and ureters dilate:
 - **Before 14 wk:** progesterone relaxes the muscularis.
 - **From midpregnancy:** ureteral compression, **especially right-sided**. Vesicoureteral reflux also occurs.
 - Consequences: ↑ **upper urinary infection** and **misread imaging**.
- Functional renal hypertrophy early; glomeruli larger (no more cells). Afferent and efferent arteriolar vasodilation.
- **GFR +20% by 12 wk;** ultimately **RPF +40%, GFR +65%** → serum creatinine and urea fall → **"normal"** **nonpregnant values may be abnormal in pregnancy**.

Assessment of renal function

Glucosuria

- Occasional glucosuria is normal (↓ tubular glucose reabsorption). Dipstick ≥1+ is linked to later GDM but has **poor sensitivity**.
- **UK (NICE): early glucose testing for an isolated 2+ or repeated 1+ — reasonable.**

Proteinuria

- Normal 24-h protein **115 mg** (95% limit **260 mg/d**), not different by trimester; albumin **5–30 mg/d**.
- **Abnormal = >300 mg/d**. Many regard **500 mg/d** as important with gestational hypertension.
- **Spot protein:creatinine ratio ≈ grams of protein per day;** best on a **first morning void**.
- Lower UTI does **not** increase proteinuria.

Hematuria

- Idiopathic dipstick hematuria (≥1+) in **3%** of nulliparas <20 wk → **2× preeclampsia** (not confirmed by all). Another series: dipstick hematuria **15%**, mostly trace, **false-positive rate 40%**.
- False positives: **vitamin C, povidone-iodine, myoglobin, free hemoglobin** → confirm with formal urinalysis.
- **Microscopic hematuria = >3 RBCs/HPF**.
- Workup: history/exam; exclude infection; glomerular disease → creatinine, BUN, PCR or 24-h protein, sediment (**dysmorphic RBCs, cellular casts**, proteinuria). **MR urography and cystoscopy** exclude tumor/stone.
- ⚠ **ACOG: no cancer workup** for asymptomatic, low-risk, never-smoking women **<50 y with ≤25 RBCs/HPF**.

Creatinine and imaging

- **Serum creatinine persistently >0.9 mg/dL (75 μmol/L) → suspect intrinsic renal disease;** creatinine clearance estimates GFR.

- **Sonography:** size, consistency, obstruction. Full IVP not routine; **one-shot pyelography** (film 30 min after contrast) if needed. **MR** for renal masses.
- Cystoscopy for usual indications; **ureteroscopy for stones ~5% complications.**

Renal biopsy

- Relatively safe but usually postponed unless it will change therapy. **Complications 7% in pregnancy vs 1% postpartum** (243 biopsies).
- Consider for **rapid unexplained decline in renal function** or **symptomatic nephrotic syndrome.**
- **~50% of normal pregnant women** have slight–moderate **glomerular endotheliosis**; all 27 with proteinuric hypertension had moderate–severe endotheliosis.

Pregnancy after unilateral nephrectomy

- The remaining kidney hypertrophies further in pregnancy; most do well.
- Check **serum creatinine and spot PCR.**
- **Prior kidney donation:** no permanent consequences, but **↑ subsequent preeclampsia.**

2. Urinary Tract Infections

- **UTIs are the most prevalent bacterial infections in pregnancy.** Perineal flora; **E. coli** causes most nonobstructive pyelonephritis.
- **~90% of these E. coli have adhesins (P- and S-fimbriae)** → ↑ adherence and virulence; maternal deaths with Dr+ and P adhesins.
- Predisposing: **urinary stasis, vesicoureteral reflux, diabetes.** Pregnancy → more severe urosepsis (reversed Th1:Th2 ratio, cytokine changes).
- **Puerperium:** ↓ bladder sensitivity (labor trauma, conduction analgesia), pain masking distention (episiotomy, laceration, hematoma), diuresis → overdistention → **catheterization → UTI.**

Asymptomatic bacteriuria (ASB)

- **Definition: $\geq 10^5$ CFU/mL of ≥ 1 species on a voided culture** (IDSA 2019), without symptoms.
- Prevalence **2–6%** (sexually active nonpregnant) and **2–7%** in pregnancy — same as nonpregnant. **Highest in African-American multiparas with sickle-cell trait**; lowest in affluent white women of low parity; ↑ with diabetes.
- Usually **already present at the first prenatal visit.** After treatment, **<1%** develop symptomatic UTI.
- Pyelonephritis can occur at **20,000–50,000/mL** → may be prudent to treat lower counts.
- **Untreated ASB → symptomatic infection in ~25%;** treatment prevents most.
- **Screen at the first prenatal visit** (AAP/ACOG, USPSTF). **Leukocyte esterase/nitrite dipstick** is cost-effective where prevalence is low; Parkland cultures (prevalence **5–8%**). Susceptibilities not needed (empirical treatment). Agar-coated dipstick culture has excellent PPV/NPV.
- Benefits beyond pyelonephritis prevention are unclear (UTI linked with LBW, preterm birth, hypertension, anemia in some studies; symptomatic UTI → **1.4× preterm delivery <37 wk**).

Treatment

- **E. coli** most common. Parkland: **nitrofurantoin macrocrystals 100 mg at bedtime × 10 days**; 100 mg bid × 7 days also satisfactory; single-dose also works.
- **⚠ Recurrence ~30% with any regimen** (may reflect covert upper tract infection).
- **First recurrence = treatment failure** → consider **21 days** of therapy. Persistent/frequent ASB → **suppression for the rest of pregnancy: nitrofurantoin 100 mg at bedtime.** Rare serious side effects: **pulmonary and hepatic toxicity.**

Table 56-1. Oral Antimicrobial Agents Used for Treatment of Pregnant Women with Asymptomatic Bacteriuria

Regimen	Agents
Single-dose treatment	Amoxicillin 3 g; ampicillin 2 g; cephalosporin 2 g; nitrofurantoin 200 mg; trimethoprim-sulfamethoxazole 320/1600 mg
3-day course	Amoxicillin 500 mg three times daily; ampicillin 250 mg four times daily; cephalosporin 250 mg four times daily; nitrofurantoin 50–100 mg four times daily or 100 mg twice daily; trimethoprim-sulfamethoxazole 160/800 mg two times daily
Other	Nitrofurantoin 100 mg four times daily for 10 days; nitrofurantoin 100 mg twice daily for 5–7 days; nitrofurantoin 100 mg at bedtime for 10 days
For treatment failures	Nitrofurantoin 100 mg four times daily for 21 days
Suppression for bacterial persistence or recurrence	Nitrofurantoin 100 mg at bedtime for remainder of pregnancy

Cystitis and urethritis

- **7.2%** of pregnant women had outpatient UTI treatment (CDC, 2014).
- **Cystitis:** dysuria, urgency, frequency; few systemic signs. Pyuria, bacteriuria, microscopic (sometimes gross — hemorrhagic cystitis) hematuria.
- **~40% of pyelonephritis is preceded by lower tract symptoms.**
- **3-day regimens ~90% effective;** single-dose less effective (exclude pyelonephritis first).
- **Pyuria + sterile culture → urethritis from *Chlamydia trachomatis*** (often with mucopurulent cervicitis) → **azithromycin.**

Acute pyelonephritis

- **Most common serious nonobstetrical medical complication of pregnancy.** ~29,000 US hospitalizations (2006); highest in **adolescents (17.5/1000)** and Hispanic women (10.1/1000). **3.5%** of antepartum admissions.
- **Leading cause of septic shock in pregnancy; 12%** of Parkland intermediate-care antepartum admissions. Maternal urosepsis may ↑ **cerebral palsy** in preterm infants. No serious long-term maternal sequelae.

Clinical findings

- More common in the **second half**; risk factors **nulliparity and young age.**
- **Right-sided >1/2; bilateral 1/4.**
- Abrupt **fever, shaking chills, lumbar pain**; anorexia, nausea, vomiting; **CVA tenderness.** Sediment: clumped leukocytes, bacteria.
- Differential: **labor, chorioamnionitis, appendicitis, abruption, infarcted leiomyoma.**
- **Bacteremia 15–20%.**

Organism	Frequency
<i>E. coli</i>	70–80%
<i>Klebsiella pneumoniae</i>	3–5%
<i>Enterobacter</i> or <i>Proteus</i>	3–5%
Gram-positive (GBS, <i>S. aureus</i>)	up to 10%

Complications (endotoxin-mediated)

- **Renal dysfunction in 5%** despite fluids (reversible) → monitor creatinine.
- **Respiratory insufficiency/pulmonary edema up to 10%** (alveolar injury) → may progress to **ARDS.**
- **⚠️ β-agonist tocolysis → pulmonary edema 8% (4×)** (Na⁺/fluid retention).
- **Uterine contractions** (related to fever): **5/h at admission → 2/h within 6 h** of fluids and antibiotics.
- **Hemolysis → anemia in ~1/3;** recovery is normal (erythropoietin unaffected).

Management

Table 56-2. Management of the Pregnant Woman with Acute Pyelonephritis

Step
Hospitalize patient
Obtain urine and blood cultures
Evaluate hemogram, serum creatinine and electrolytes
Monitor vital signs frequently, including urinary output — consider indwelling catheter
Establish urinary output ≥ 50 mL/h with IV crystalloid
Administer IV antimicrobial therapy (see text)
Obtain chest radiograph if dyspnea or tachypnea
Repeat hematology and chemistry studies in 48 h
Change to oral antimicrobials when afebrile
Discharge when afebrile 24 h ; consider antimicrobial therapy for 7–10 d
Repeat urine culture 1–2 wk after antimicrobial therapy completed

IV = intravenous. From Lucas, 1994; Sheffield, 2005. (The text recommends completing 10–14 days of total therapy; the table says 7–10 days of oral therapy after discharge.)

- Blood cultures (Parkland: if **$>39^{\circ}\text{C}$**) have limited utility.
- **IV hydration = cornerstone.** Antibiotics promptly — may transiently **worsen endotoxemia** (bacterial lysis). Monitor urine output, BP, pulse, temperature, O_2 saturation.
- Lower high fever (**cooling blanket, acetaminophen**) — especially early in pregnancy (hyperthermia teratogenic).
- **Antibiotics** (all ~95% effective): **ampicillin + gentamicin** (Parkland first-line); **cefazolin or ceftriaxone**; or an extended-spectrum agent. **$<1/2$ of *E. coli* are ampicillin-sensitive**, but gentamicin + ampicillin/cephalosporin is synergistic. Monitor creatinine with nephrotoxic drugs.
- **95% afebrile by 72 h.** After discharge, oral therapy to complete **10–14 days**.
- **Recurrence 30–40%** → **nitrofurantoin 100 mg at bedtime for the rest of pregnancy**.

Persistent infection

- Normal response: defervescence **$\sim 1^{\circ}\text{F}$ per day**; **$\sim 20\%$ febrile >4 days**.
- **⚠ No improvement by 48–72 h** → **look for obstruction, calculi or abscess** → **renal sonography**.
 - Stones may be missed by US → **plain film detects $\sim 90\%$** ; or modified one-shot IVP.
 - **Intrarenal/perinephric abscess or phlegmon** → **MR** if US inconclusive.
- **Relieve obstruction: percutaneous nephrostomy** preferred (easier to replace) or cystoscopic **double-J stent** (left until after delivery; encrusts, may need replacing). Some need stone surgery.
- **Abscess >3 cm** → percutaneous drainage or surgery.

Outpatient management

- 92 women given IM ceftriaxone 1 g $\times$ 2 (24 h apart): only **$1/3$ suitable** for outpatient care, and **$1/3$ of those failed** → **outpatient management suits very few pregnant women**.

Reflux nephropathy

- Nephron loss from patchy interstitial scarring and tubular atrophy due to **high-pressure reflux of sterile urine**.
- Long-term **hypertension** (severe if renal damage). Impaired function and **bilateral scarring** → **more hypertensive disorders** (879 pregnancies).
- Women with childhood reflux surgery commonly have **bacteriuria** in pregnancy.

3. Nephrolithiasis

- Lifetime risk **9%** in women; onset in the **3rd decade**.
- **Calcium salts ~80%** → exclude **hyperparathyroidism**. Nonpregnant young women: **calcium oxalate**; **pregnancy: 65–75% calcium phosphate/hydroxyapatite** (↑ urinary pH ↓ calcium phosphate solubility).
- More common in parous women. **>½ form another stone within 10 years**.
- ⚠ **A low-calcium diet promotes stones**. Prevention: **hydration, low-sodium, low-protein diet; thiazides** ↓ stones.
- **Indications for removal: obstruction, infection, intractable pain, heavy bleeding**. Cystoscopic basket extraction reasonable; lithotripsy not generally recommended in pregnancy.

Stone disease during pregnancy

- Incidence **0.3–0.8 per 1000** pregnancies; mostly **2nd and 3rd trimesters**. Bladder stones rare (may obstruct labor).
- Outcomes conflicting: preterm delivery **10.6% vs 6.4%**; Taiwan **20–40% ↑ LBW and preterm**; others no difference. Stone formers before pregnancy → **↑ GDM and preeclampsia**.

Diagnosis

- **Pain in >90%** (possibly fewer symptoms due to dilation). **Gross hematuria 2–23%** (less than nonpregnant).
- **Sonography first-line**, but many stones missed. **Indirect signs: abnormal dilation or absent ureteral "jets"**. Dilation without a visible stone → one-shot IVP.
- **MR = second-line** after nondiagnostic US. If CT needed: **unenhanced helical CT, fetal dose ~7 mGy**.

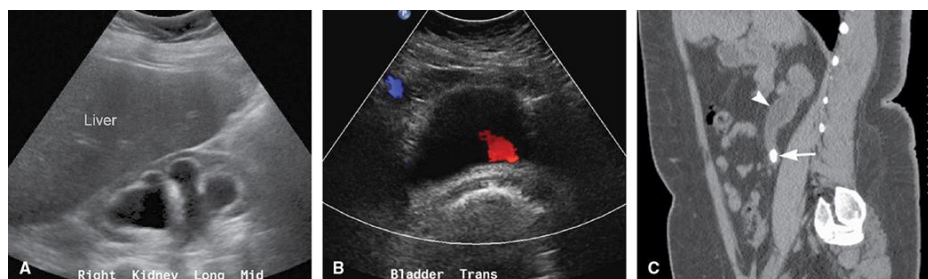


Figure 56-1. Obstructing stone: (A) severe right hydronephrosis with thin parenchyma; (B) color Doppler shows a left ureteral jet but none from the right; (C) CT shows the mid-right ureteral stone (arrow) with hydroureteronephrosis.

Management

- **IV hydration and analgesia**; strain urine. **Infection in ½** of symptomatic stones → treat vigorously. Persistent pyelonephritis → search for obstructing stone.
- **~75% improve with conservative therapy** and pass the stone.
- Others: **ureteral stent, ureteroscopy, percutaneous nephrostomy, transureteral laser lithotripsy** (holmium:YAG successful in 9/10), **basket extraction**. **<2%** need surgical exploration.
- Percutaneous nephrolithotomy limited by fluoroscopy. ⚠ **Extracorporeal shock-wave lithotripsy is contraindicated in pregnancy**.

4. Pregnancy After Renal Transplantation

- >100,000 on the US waiting list (2020). **1-year graft survival 95% (living donor), 90% (deceased donor)**.
- Immunosuppression: cyclosporine, OKT3 (anti-CD3), **tacrolimus**; ⚠ **mycophenolate mofetil is teratogenic — 23% birth defects**.
- **Fertility returns promptly** after transplant → **counsel on contraception** (esp. on mycophenolate); >½ report no counseling.

Pregnancy outcomes

- ↑ Severe maternal morbidity. >4700 pregnancies: **~70% live births**; ↑ preeclampsia, GDM, preterm delivery.
- UK cohort: live births **>90%**; **½ delivered <37 wk** (only **9% <32 wk**); **LBW ½, FGR 25%**; malformations not ↑ (except mycophenolate).
- **Preeclampsia 22%** (UK; up to ⅓ in reviews). ⚠ **Rejection can be hard to distinguish from preeclampsia**; rejection only **~2%**.
- **BK virus (polyomavirus hominis 1)** infections frequent; **GDM ~5%** (immunosuppression).
- **Cyclosporine levels fall** with volume expansion (not linked to rejection). Cyclosporine and tacrolimus are **nephrotoxic** and cause hypertension; cyclosporine enters **breast milk**. Offspring neurodevelopment normal in small studies; late effects (malignancy, germ cells) remain a concern.
- Hyperfiltration did not harm long-term graft survival (34 recipients, 15 y).

Italian Society of Nephrology — prerequisites before pregnancy

Requirement	Criterion
General health	Good for ≥1–2 years after transplant
Stable renal function	GFR ≥60 mL/min, creatinine <2 mg/dL, proteinuria <500 mg/d
Rejection / drugs	No recent rejection ; no teratogenic drugs
Hypertension	Absent or well controlled

Management

- **Close surveillance.** Treat covert bacteriuria; suppress if recurrent.
- Serial **liver enzymes and blood counts** (azathioprine, cyclosporine toxicity); ± cyclosporine levels.
- On **corticosteroids** → exclude overt diabetes with **GTT at first visit**.
- Surveillance for **HSV, CMV, toxoplasmosis**; BK virus in known carriers.
- **GFR usually ↑ 20–25%**. Significant creatinine rise → differential: **acute rejection, cyclosporine toxicity, preeclampsia, infection, obstruction**. Pyelonephritis or rejection → admit; imaging ± biopsy.
- Watch for hypertension and **superimposed preeclampsia** (manage as Ch 53). Fetal surveillance for FGR and preterm birth.
- **Cesarean for obstetric indications** (occasionally the transplanted kidney obstructs labor), yet overall cesarean rate **~80%**.

5. Polycystic Kidney Disease (PKD)

- Usually **autosomal dominant ciliopathy** of renal tubules. **~1 in 1000**; causes **5–10% of US ESRD**.
- **PKD1 (chromosome 16) ~85%; PKD2 (chromosome 4) ~15%**. Prenatal diagnosis if familial mutation/linkage known; **PGT ↓ affected fetuses to 1–2%**.
- Worse in men; symptoms in **3rd–4th decade**: flank pain, hematuria, proteinuria, masses, stones, infection. **Hypertension 75%**; progression to renal failure; acute failure from infection or **ureteral angulation** by cysts.
- Extrarenal: **hepatic cysts ⅓** (more aggressive in women); **~10% die from ruptured intracranial berry aneurysm**; **valvular lesions up to ¼** (MVP; mitral, aortic, tricuspid incompetence).

Pregnancy outcomes

- Depend on **hypertension and renal insufficiency**. **UTIs common**.
- 605 pregnancies: composite perinatal complications **33% vs 26%** (similar), but **more hypertension/preeclampsia**. Pregnancy does **not accelerate** the disease.

6. Glomerular Diseases

- Glomerulonephritis = capillary inflammation, often autoimmune; single stimulus (post-streptococcal) or systemic (SLE, diabetes). Persistent disease → progressive renal decline, often silent until CKD.
- Goals before pregnancy: **stabilize disease, minimize proteinuria (immunosuppression), control hypertension, delay pregnancy until achieved.**

Table 56-3. Patterns of Clinical Glomerulonephritis

Syndrome	Causes
Acute nephritic syndromes	Poststreptococcal, infective endocarditis, SLE, antiglomerular basement membrane disease, IgA nephropathy , ANCA vasculitis, Henoch-Schönlein purpura, cryoglobulinemia, membranoproliferative and mesangioproliferative GN
Pulmonary-renal syndromes	Goodpasture , ANCA vasculitis, Henoch-Schönlein purpura, cryoglobulinemia
Nephrotic syndromes	Minimal change disease, FSGS, membranous GN, diabetes, amyloidosis, others
Basement membrane syndromes	Antiglomerular basement membrane disease, others
Glomerular vascular syndromes	Atherosclerosis, chronic hypertension, sickle-cell disease, thrombotic microangiopathies, antiphospholipid antibody syndrome, ANCA vasculitis, others
Infectious disease-associated syndromes	Poststreptococcal, infective endocarditis, HIV, HBV, HCV, syphilis, others

ANCA = antineutrophilic cytoplasmic antibodies; HBV = hepatitis B virus; HCV = hepatitis C virus; HIV = human immunodeficiency virus; IgA = immunoglobulin A; SLE = systemic lupus erythematosus.

Acute nephritic syndromes

- **Hypertension, hematuria, red-cell casts, pyuria, proteinuria**; salt/water retention → edema, circulatory congestion.
- May resolve, become **rapidly progressive** (→ ESRD) or chronic.
- **Poststreptococcal GN** = prototype (confused with eclampsia until the mid-1800s).
- **IgA nephropathy (Berger disease) = most common acute GN worldwide**; systemic form = Henoch-Schönlein purpura.
- **Anti-GBM (IgG) disease**; with lung involvement (alveolar hemorrhage) = **Goodpasture syndrome**.

Pregnancy

- Hard to distinguish from **severe preeclampsia** (e.g., lupus flare in the 2nd half) → renal biopsy sometimes needed.
- Primary GN (395 pregnancies, mostly normal function): **½ hypertension, ¼ preterm, perinatal mortality 80/1000 after 28 wk**; worst with impaired function, early/severe hypertension, nephrotic proteinuria.
- 48 pregnancies: **⅓ preeclampsia**, 39% doubled proteinuria, 27% creatinine ↑ ≥50%, **½ preterm, 13% perinatal death**.
- **IgA nephropathy** (376 pregnancies): perinatal mortality 12%, preterm 9%, LBW 10%, preeclampsia 7%. Whether pregnancy worsens IgA nephropathy is disputed.

Nephrotic syndromes

- **Heavy proteinuria + hypoalbuminemia + hypercholesterolemia + edema**; hypertension often; albumin nephrotoxicity → progressive insufficiency.

Table 56-4. Causes of the Nephrotic Syndrome in Adults

Cause	Features
Focal segmental glomerulosclerosis (FSGS) (35%)	Viruses, hypertension, reflux nephropathy, sickle-cell disease
Membranous glomerulonephritis (~20%)	Idiopathic (most cases), malignancy, infection, connective tissue diseases
Minimal change disease (MCD) (10–15%)	Primary idiopathic (most cases), drug-induced (NSAIDs), allergies, viral infections
Diabetic nephropathy	Most common cause of ESRD
Glomerular deposition diseases	Light-chain, amyloidosis

ESRD = end-stage renal disease; NSAIDs = nonsteroidal antiinflammatory drugs.

Management

- Treat by etiology (biopsy helps). **Normal amounts of high-biological-value protein** (high-protein diets ↑ proteinuria).
- **⚠ Thromboembolism ↑** (esp. **renal vein thrombosis**; arterial too): platelet hyperactivity, **urinary loss of antithrombotic factors**, ↑ hepatic prothrombotic factors.
- **Thromboprophylaxis throughout pregnancy if serum albumin <2 g/dL or proteinuria >3.5 g/24 h** (some recommend).
- Primary glomerular disease may respond to steroids/immunosuppressants; infection- or drug-induced cases remit when the cause is removed.

Pregnancy

- **½** with nephrotic-range proteinuria have rising protein excretion; Parkland: **⅔ exceeded 3 g/d**. With mild dysfunction, GFR still augments normally.
- Edema worsened by leg hydrostatic pressure; **massive vulvar edema** with comorbidities (diabetes, syphilis, preeclampsia).
- **Up to ½ have chronic hypertension**; preeclampsia is common and early.
- Without severe hypertension or renal insufficiency → usually successful; with them → much worse.
- **Parkland (65 pregnancies):** protein **~4 g/d**; renal insufficiency **75%**, chronic hypertension **40%**, anemia **25%**, preeclampsia **60%**, preterm **45%**; yet **53/57** liveborn.



Figure 56-2. Massive vulvar edema in a gravida with heavy proteinuria from preeclampsia.

Long-term

- **≥20% progress to ESRD within 10 years** (Parkland). Shortest renal survival with **creatinine >1.4 mg/dL and proteinuria >1 g/d**.

7. Chronic Kidney Disease (CKD)

- NKF staging by GFR: best stage **GFR ≥ 90 mL/min/1.73 m² → stage 5 GFR < 15 .**
- Commonest causes of ESRD: **diabetes, chronic hypertension, glomerulonephritis, PKD.**
- **Outcome depends mainly on degree of renal impairment and hypertension**, not the specific disease; systemic disease (diabetes, SLE) worsens prognosis.

Older functional category (Davison)	Serum creatinine
Normal or mild impairment	< 1.5 mg/dL
Moderate impairment	1.5–3.0 mg/dL
Severe insufficiency	> 3.0 mg/dL

The text describes six NKF stages "from stage 0 (GFR ≥ 90)"; the NKF numbers stages 1–5 (stage 1 GFR ≥ 90), with stage 3 split into 3a/3b.

Pregnancy and CKD

- High rates of hypertension/preeclampsia, preterm birth, FGR. **Low-dose aspirin from 12–28 wk** to ↓ preeclampsia.
- Nephron loss → hyperfiltration of survivors → nephrosclerosis.
- **Pregnancy augmentation of RPF/GFR:** preserved with mild insufficiency; **only ½ with moderate** insufficiency; **none with severe.**
- **Blood volume expansion correlates inversely with creatinine:** mild–moderate **~55%**; severe **~25%** (like eclampsia) → plus chronic renal anemia → **hypovolemia and anemia even with normal delivery blood loss.**

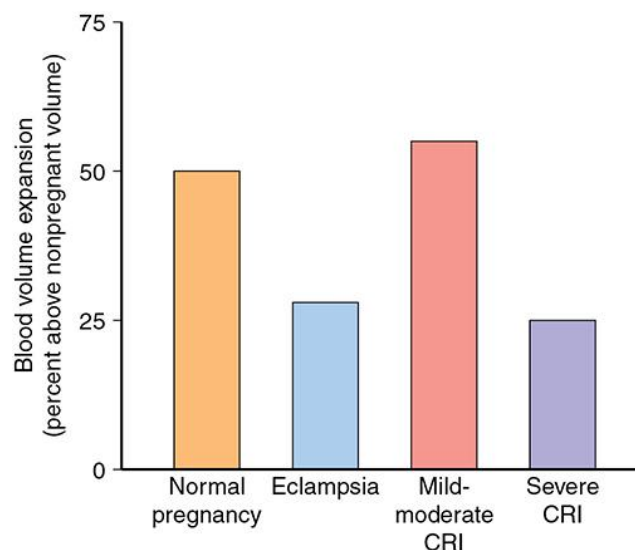


Figure 56-3. Blood volume expansion: normal pregnancy ~50%, mild–moderate CRI ~55%, but only ~25% with severe CRI — similar to eclampsia (~28%).

Renal disease with preserved function

- Still ↑ complications (Table 56-5). Biopsy-proven glomerular disease (123 pregnancies): **40%** obstetric or renal complications.
- GN with minimal insufficiency (395): impaired function **15%**, worse proteinuria **60%**, hypertension **$> 1/2$** (only 12% chronic HTN beforehand); **perinatal mortality 140/1000** (50/1000 even without severe features); **permanent decline in 5%.**

Table 56-5. Complications (%) Associated with Chronic Kidney Disease During Pregnancy

Complication	Preserved Function	Renal Insufficiency: Moderate and Severe	Renal Insufficiency: Severe
Chronic hypertension	25	30–70	50
Gestational hypertension	20–50	30–50	75
Worsening renal function	8–15	20–43	35
Permanent dysfunction	4–5	10–20	35
Preterm delivery	7	30–60	73
Fetal-growth restriction	8–14	30–38	57
Perinatal mortality	5–14	4–7	0

Data from Alsuwaida, 2011; Cunningham, 1990; Farwell, 2013; He, 2018; Imbasciati, 2007; Maruotti, 2012; Nevis, 2011; Piccoli, 2010, 2011; Stettler, 1992; Trevisan, 2004; Zhang, 2015.

Chronic renal insufficiency

- Complications rise with impairment. Systematic review (1514 CKD pregnancies) — odds ratios: **preeclampsia 10.4, preterm 6.3, FGR 4.9, cesarean 2.7**.
- Despite frequent hypertension, anemia, preeclampsia, preterm birth and FGR, perinatal outcomes generally acceptable.

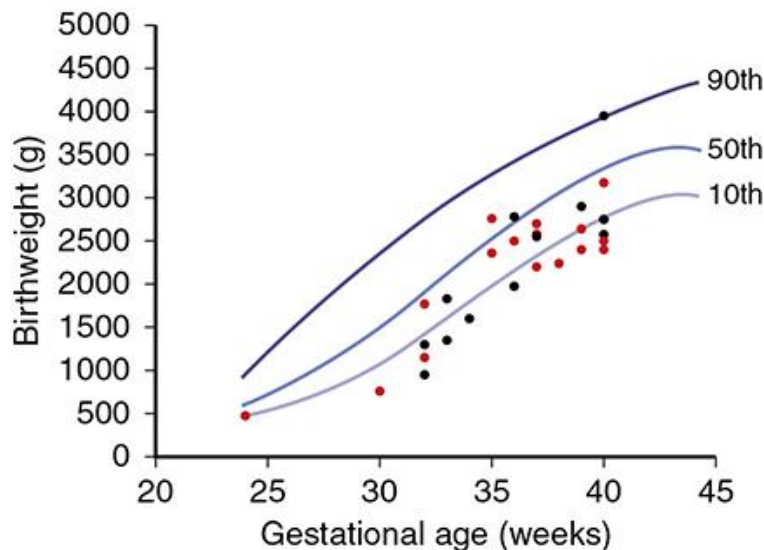


Figure 56-4. Birthweights in renal insufficiency cluster at or below the 10th percentile, more so with severe disease (red).

Management

- **Frequent BP monitoring**; creatinine and 24-h protein as indicated.
- **Treat bacteriuria** (↓ pyelonephritis and nephron loss).
- **⚠ Protein-restricted diets not recommended.**
- Anemia may respond to **recombinant erythropoietin** (side effect: hypertension).
- **Serial sonography** for growth.
- Worsening hypertension vs superimposed preeclampsia is hard to separate; **PIGF and sFlt-1** may help.

Long-term effects

- Pregnancy may accelerate CKD (hyperfiltration → nephrosclerosis), especially with severe insufficiency; few long-term effects with normal baseline function (360 women with GN).
- **Moderate–severe insufficiency: 20% ESRD by ~4 years** (Parkland); confirmed in stage 3–4 CKD.
- **10% ESRD at 1 year** postpartum (Jones & Hayslett). Prepregnancy **creatinine ≥1.4 mg/dL + proteinuria >1 g/d** → **30% ESRD at ~3 years**.

- **Chronic proteinuria found in pregnancy → 20% ESRD within several years.**
- But nonpregnant women lose function at the same rate → pregnancy may not hasten a predetermined decline.

Dialysis during pregnancy

- Severe renal impairment → **subfertility**, correctable with hemodialysis or peritoneal dialysis.
- Maternal complications: **severe hypertension, abruption, heart failure, sepsis.**

Table 56-6. Pregnancy Outcomes in Women Undergoing Dialysis During Pregnancy

Study (Year)	N	Delivery (weeks)	Birthweight (g)	Hypertension (%)	Hydramnios (%)	Perinatal Mortality (%)	Surviving Infants (%)
Toma (1999)	54	31.9	1545	35	44	33	67
Chao (2002)	13	32	1540	72	46	31	69
Tan (2006)	11	31	1390	36	18	18	82
Chou (2008)	13	30.8	1510	57	71	50	50
Luders (2010)	52	32.7	1555	67	40	13	87
Shahir (2013)	13	NS	2130	19 ^a	14	22	78
Jesudason (2014)	77	33.8	1750	NS	NS	20	80
Approximate averages	233	~32	~1600	~50	~44	~25	~75

^aPreeclampsia only. NS = not stated.

- **Perinatal mortality ~250/1000** (Table 56-6); systematic review (681 pregnancies): **180/1000. More dialysis hours/week → better outcomes; hemodialysis superior to peritoneal dialysis.**
- Already on dialysis → continue the same method, **increasing frequency.**
- **Start dialysis when creatinine clearance ≤20 mL/min** (consider). **5–6 sessions/week** to avoid abrupt volume shifts and hypotension.
- **Protocol: double multivitamins**; oral **calcium and iron**; adequate protein and calories; **erythropoietin** for anemia; dialysate with **extra calcium and less bicarbonate.**

8. Acute Kidney Injury (AKI)

- **Definition:** creatinine ↑ **≥0.3 mg/dL within 48 h**, or **≥50% from baseline within a week**, or urine output **<0.5 mL/kg/h for >6 h**. Pregnancy criteria not standardized (most use ↑ ≥0.3 mg/dL).
- Incidence **2.68 per 10,000 births** (Canada, 2009–2010), much lower than before.
- Older series (266 women): **preeclampsia ~70%**, obstetric **hemorrhage ~50%**, **abruption ~30%**; **dialysis ~20%**; **maternal mortality ~15%**.
- Main causes today: **severe preeclampsia and hemorrhage** (abruption, **HELLP**), **sepsis** (resource-poor settings), **acute fatty liver** (renal insufficiency in virtually all 52 Parkland cases), **hyperemesis** dehydration, **thrombotic microangiopathies.**

Diagnosis and management

- Mostly **postpartum** (no fetal issues). Acute creatinine rise is usually **renal ischemia**; oliguria is also key. Prerenal and intrarenal factors often coexist (e.g., abruption hypovolemia + preeclampsia).
- Severe azotemia and persistent oliguria → **hemofiltration or dialysis early**, before marked deterioration (↓ mortality, better recovery). Normalize hemodynamics.
- **Adjust drug doses** — especially **magnesium sulfate.**
- Function usually returns to (near) normal, but later pregnancies have ↑ hypertension and adverse fetal outcomes.

Prevention

1. **Prompt, vigorous volume replacement** (crystalloid + blood) for massive hemorrhage — abruption, previa, uterine rupture, atony
2. **Deliver/terminate** pregnancies with severe preeclampsia or eclampsia; careful transfusion if blood loss is above average
3. **Close watch for sepsis and shock** with pyelonephritis, septic abortion, chorioamnionitis, pelvic infection
4. **⚠ No loop diuretics for oliguria** until blood volume and cardiac output are adequate
5. **Vasoconstrictors judiciously**, only once pathological vasodilation is confirmed as the cause of hypotension

Acute cortical necrosis

- Irreversible ischemic failure; now **exceedingly uncommon** (formerly $\frac{1}{4}$ of obstetric renal failure). Followed abruption, preeclampsia–eclampsia, endotoxic shock (septic abortion).
- Thrombosis of renal vascular segments (focal → gross). Indistinguishable from ATN early; recovery variable; may leave stable insufficiency.

Obstructive renal failure

- Rarely, a **very large uterus** compresses both ureters → oliguria and azotemia.
- Twins at 34 wk: anuria, creatinine **12.2 mg/dL** → after **amniotomy**, urine **500 mL/h** and creatinine normalized.
- **Ureteral stents** (10 women) left a mean **15.5 wk**, removed **4–6 wk postpartum**.
- ↑ Risk after **prior reflux surgery**. Partial obstruction → fluid retention and hypertension; relief → diuresis. Hydramnios (9.4 L) case: amnioreduction → diuresis, ↓ creatinine, better BP.

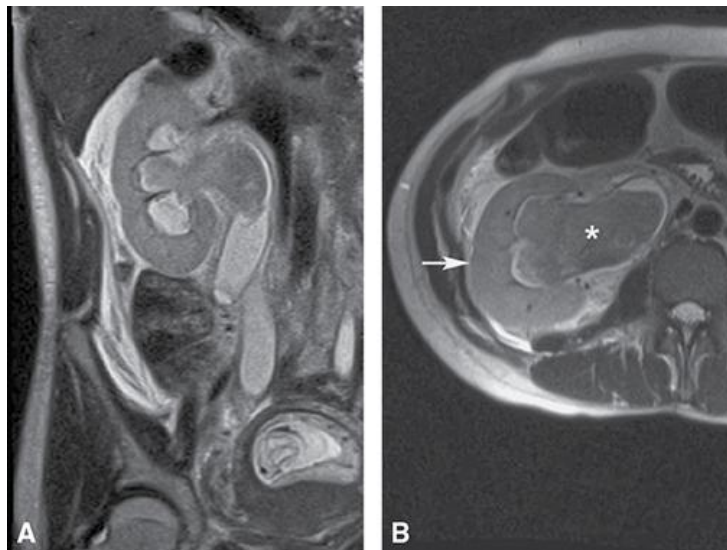


Figure 56-5. MR of ureteral obstruction with hydronephrosis in pregnancy (creatinine 8 mg/dL → 0.8 mg/dL after percutaneous nephrostomy).

9. Lower Urinary Tract Lesions

Urethral diverticulum

- From a **paraurethral gland abscess** rupturing into the urethra → persistent sac with ostium.
- Urine dribbling from the sac, pain, **palpable mass**, **recurrent UTIs**.
- **Managed expectantly in pregnancy**; rarely drainage or surgery. **MR** preferred for further evaluation.

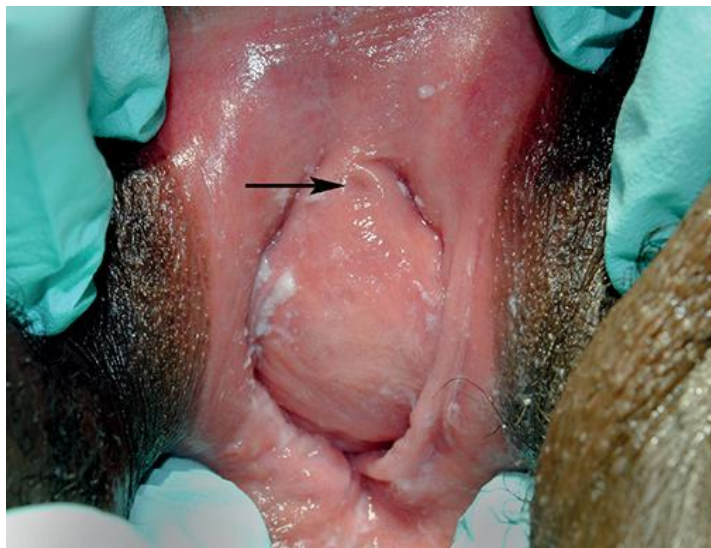


Figure 56-6. Asymptomatic urethral diverticulum: midline bulge beneath the urethral meatus (arrow).

Urinary tract fistulas

- Usually pre-existing; rarely form in pregnancy.
- **Vesicovaginal fistula after McDonald cerclage; prolonged obstructed labor** (resource-poor countries) → pressure necrosis between fetal head and pelvis.
- **Vesicouterine fistula** after prior cesarean; **vesicocervical fistula** after cesarean or compression of the anterior cervical lip against the symphysis.

Quick-Recall: Urinary Tract Infection Spectrum in Pregnancy

	Asymptomatic bacteriuria	Cystitis	Acute pyelonephritis
Definition / features	$\geq 10^5$ CFU/mL, no symptoms	Dysuria, urgency, frequency; few systemic signs	Fever, chills, flank pain, CVA tenderness; bacteremia 15–20%
Frequency	2–7%	7.2% treated as outpatients	Most common serious nonobstetric complication
Why it matters	25% → symptomatic if untreated	40% of pyelonephritis preceded by lower tract symptoms	Septic shock, ARDS (up to 10%), renal dysfunction (5%), anemia ($\frac{1}{3}$), contractions
Treatment	Nitrofurantoin 100 mg qHS $\times$ 10 d (or Table 56-1)	3-day regimens (~90% effective)	Admit, IV fluids (UO ≥ 50 mL/h), ampicillin + gentamicin or cephalosporin; 95% afebrile by 72 h
Recurrence	~30% → 21 days, then suppression	—	30–40% → nitrofurantoin 100 mg qHS rest of pregnancy
Red flag	—	Pyuria + sterile culture → <i>Chlamydia</i>	No response by 48–72 h → US for obstruction/stone/abscess

Key Numbers to Memorize

- GFR **+20%** by 12 wk; ultimately RPF **+40%**, GFR **+65%**; creatinine **>0.9 mg/dL** abnormal
- Proteinuria normal **115 mg/d** (limit 260); abnormal **>300 mg/d**; PCR $\approx$ g/day
- Microscopic hematuria **>3 RBC/HPF**; no cancer workup if <50 y, never-smoker, **≤ 25 RBC/HPF**
- Renal biopsy complications **7%** pregnant vs **1%** postpartum
- ASB $\geq 10^5$ CFU/mL, **2–7%**; untreated → **25%** symptomatic; recurrence **30%**

- Pyelonephritis: *E. coli* **70–80%**; bacteremia **15–20%**; right-sided **>1/2**; renal dysfunction **5%**; pulmonary insufficiency **≤10%** (β -agonists **8%**); anemia **1/3**; **95%** afebrile by **72 h**; recurrence **30–40%**; UO **≥50 mL/h**
- Stones: lifetime **9%**; pregnancy **0.3–0.8/1000**; **65–75%** calcium phosphate; **75%** pass spontaneously; plain film sees **~90%**; CT fetal dose **~7 mGy**; **no ESWL**
- Transplant: wait **1–2 y**; GFR **≥60**, creatinine **<2**, proteinuria **<500 mg/d**; preeclampsia **22%**; rejection **~2%**; mycophenolate **23%** defects; cesarean **~80%**
- PKD: **1/1000**; PKD1 **85%**; hypertension **75%**; berry aneurysm deaths **10%**
- Nephrotic prophylaxis: albumin **<2 g/dL** or proteinuria **>3.5 g/d**; preeclampsia **60%**; **≥20%** ESRD in 10 y
- CKD categories: creatinine **<1.5 / 1.5–3.0 / >3.0**; blood volume **+55%** (mild–moderate) vs **+25%** (severe)
- CKD ORs: preeclampsia **10.4**, preterm **6.3**, FGR **4.9**, cesarean **2.7**
- Dialysis: start at CrCl **≤20 mL/min**; **5–6×/wk**; perinatal mortality **~250/1000** (180 in review); delivery **~32 wk**, **~1600 g**
- AKI: creatinine $\uparrow$ **≥0.3 mg/dL/48 h** or **≥50%/wk**, UO **<0.5 mL/kg/h × 6 h**; **2.68/10,000**; historical mortality **~15%**